

INVESTMENT & TECHNOLOGY OVERVIEW

BioTC: Turn Sludge Liability into Clean Energy

1. Executive Summary

BioTC (developed by BioTheCon and represented by **BTC Consulting Sp. z o.o.**) is an integrated clean-tech solution that solves the final, most critical bottleneck of wastewater treatment: **sewage sludge and organic waste accumulation**.

By combining **Thermal Hydrolysis (TH)** and **Hydrothermal Carbonization (HTC)** in a single, closed-loop system, BioTC converts wet, hazardous organic sludge into sterile, high-energy **Eco-Coal** and **clean, pathogen-free water** on-site.

WASTE REDUCTION

Sludge volume is compressed by up to **80% (5x volume reduction)**, removing the water component and drastically decreasing logistical costs.

ENERGY PRODUCTION

Converts wet feedstock into dry **Eco-Coal** biofuel (20–25 MJ/kg), usable in industrial boilers for heating or sold on the market.

2. The Clean-Water Paradox & Regulatory Pressure

Wastewater treatment plants (WWTPs) excel at returning clean water to rivers. However, this process filters out a toxic, wet, pathogen-ridden residue: **sewage sludge**. Current management methods are facing critical challenges:

The Logistic & Financial Strain

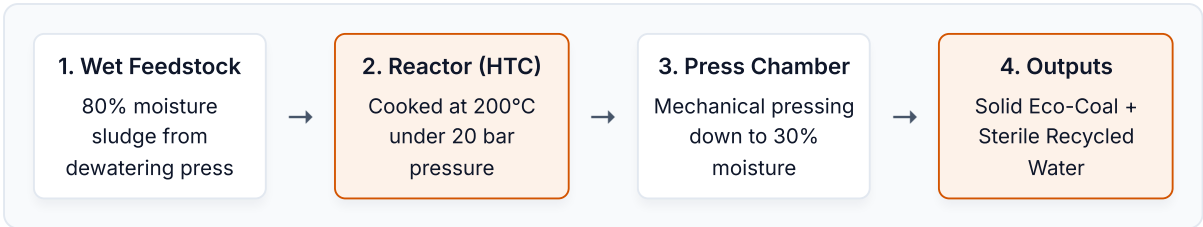
Sewage sludge consists of **80% moisture**. Traditional hauling and landfill disposal are expensive because operators are essentially paying to transport water. Environmental taxes and landfill dumping fees are rising exponentially, making old disposal methods unsustainable.

Tighter EU Directives

Under new EU regulations (including ****Directive 2024/3019****), agricultural spreading and landfilling of raw sludge are being banned. Microplastics, pharmaceutical traces, and PFAS ("forever chemicals") must be completely destroyed to prevent soil and groundwater contamination.

3. The Technology: Integrated TH + HTC

Instead of relying on energy-intensive air-drying or incineration, BioTC uses a pressurized thermochemical process that mimics natural coal formation over millions of years, compressing it into a **2-hour cycle**.



- **Thermal Destruction:** High-pressure cooking at 180–220 °C breaks down cellular walls and completely destroys 100% of pathogen structures, bacteria, microplastics, hormone residues, and PFAS.
- **Physical Phase Change:** The sludge loses its colloidal structure. Water, which was chemically bound inside sludge cells, is released and can be easily pressed out mechanically without high heat.
- **Digestion Booster (HTC-D):** The liquid run-off after mechanical press is rich in soluble organics. Returning it back to anaerobic digestion tanks stimulates additional ****biogas yields by up to 60%****.

4. Value-Added Outputs

The BioTC process converts a toxic municipal liability into two valuable commodities:

● **Eco-Coal (Hydrochar)**

A dry, sterile, and odorless solid carbon fuel with high heating density:

- **Energy Density:** 20–25 MJ/kg (equivalent to coal).
- **CO2 Neutrality:** Classified as sustainable biofuel.
- **Uses:** Combustion in boilers, cement plant feedstock, soil amendment.

● **Clean Recycled Water**

All contaminants are neutralized by the high-pressure steam:

- **Safety:** Free from PFAS, microplastics, and active bacteria.
- **Recycling:** Safely returned to the WWTP inlet.
- **Volume:** Fully sterilised water run-off.

5. Financial & Investment Case (ROI)

Implementing BioTC transitions sewage management from an **annual budget expense** to a **profitable revenue stream** by generating fuel on-site.

Feature	Traditional Landfilling	Competitor Drying/THP	BioTC Integrated TH+HTC
Sludge Volume Reduction	0%	Up to 50%	Up to 80% (5x reduction)
PFAS & PFAS Destruction	No	Partial / Weak	100% Destroyed
Energy Profile	High Cost (Transport)	Self-consuming (heat only)	Net-Positive (electricity + heat)
Biogas Yield Boost	0%	+30%	+60% (boosts digestion)
Financial Payback	Infinite Cost	High CAPEX / Slow ROI	4–6 Years Return on Investment

Green Grant Funding: Because BioTC meets strict circular economy criteria, municipal projects are eligible for European subsidies and green grants covering **up to 70% of capital expenditure (CAPEX)**.

6. Proven Credentials & Partners

BioTC is backed by leading European institutions, removing the risk of prototype testing:

- The Lubin Milestone (Poland):** In October 2025, the municipal water utility **MPWiK Lubin** officially approved and purchased the copyrights to implement BioTheCon’s integrated TH-AD-HTC concept for their municipal modernization.
- Academic Partners:** Developed in close collaboration with thermal and environmental engineering professors at the **AGH University of Krakow**, supported by national research grants.
- Industrial General Contractor:** All installations are engineered and built under general contract by **INTROL Group**, a major European engineering holding listed on the Warsaw Stock Exchange.

7. Request a Free Waste Audit

Before launching a project, we calculate the exact business case using your specific material. We offer a **Free Organic Feedstock Audit**:

- Send a 2 kg sample of your municipal or industrial sludge/feedstock to our laboratory.

- Our lab in Poland will process it in an HTC test reactor to measure the exact **Eco-Coal yield** (MJ/kg).
- We will provide a tailored **financial payback simulation (ROI)** report for your facility.